Joker of Academics

Multi-Agent Simulation of Pre-Submission Academic Behaviors: A 16-Type MBTI Personality-Based Multimodal Analysis of Graduate Thesis Completion Patterns

Dr. Water Bear[1]

[1] Department of PUA, Multiverse University

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Abstract

This study presents a comprehensive multi-agent simulation framework to investigate the behavioral patterns of graduate students during the critical pre-submission phase of thesis completion. By integrating the Myers-Briggs Type Indicator (MBTI) 16-type personality model with agent-based modeling (ABM), we simulated 3,200 agents across 20 replications to examine how personality traits influence time management, spatial preferences, economic expenditure, attention allocation, and ultimate thesis quality. Our results reveal significant associations between personality dimensions and academic performance: Thinking (T) and Judging (J) traits positively predict thesis quality (beta = 0.55, $p < 0.001$ and beta = 0.63, $p < 0.001$ respectively), while Extraversion (E) correlates with higher expenditure but lower attention maintenance. The simulation demonstrates distinct spatiotemporal patterns across the four MBTI families (Analysts, Diplomats, Sentinels, Explorers), with Analysts showing superior data analysis capabilities and Sentinels demonstrating optimal time management. Principal component analysis confirms that MBTI types form distinct behavioral clusters, validating the model's predictive validity. These findings provide evidence-based insights for personalized graduate education interventions and academic advising strategies.

Keywords: Agent-based modeling; MBTI; Graduate student behavior; Multimodal analysis; Academic performance simulation; Personality psychology

1. Introduction

In the framework of **Academic Relativity**, time flow is not a uniform scalar but a function of the Euclidean distance between the researcher and the **Deadline** (D). When D is large, the system remains in a "low-gravity" state characterized by **Academic Inertia**, where time feels thin and is dissipated through mindless browsing and logistics tracking.

However, as the coordinate rapidly collapses toward D, the intensifying **"Task-Gravity"** warps the Spacetime fabric. Observation shows that while the final hour before a deadline strictly maintains its 3600-second physical span, its **Information Flux** and **Byte Output** statistically outweigh the previous 30 days combined.

This phenomenon is defined here as **Lorentz-Academic Contraction**: a critical interval where neural synapses enter a **Superconducting State**, forcing the completion of a months-long logical chain within mere seconds.

The thesis completion process involves complex decision-making at multiple levels: daily choices about time allocation, weekly planning of research activities, strategic decisions about methodology and scope, and ongoing adjustments in response to feedback and emerging findings. These decisions are not made in isolation but are shaped by individual characteristics, environmental constraints, and social influences. Previous research has identified various factors influencing academic performance, including study habits, motivation, prior achievement, and environmental conditions. However, individual differences in personality traits have received limited attention in the context of thesis completion behaviors, despite their potential to explain significant variance in academic outcomes and their practical utility for personalized educational interventions.

Personality psychology offers a robust theoretical framework for understanding individual differences in academic behavior. The Myers-Briggs Type Indicator (MBTI), based on Carl Jung's theory of psychological types and developed by Katharine Cook Briggs and Isabel Briggs Myers, categorizes individuals into 16 distinct personality types across four dichotomies:

Extraversion/Introversion (E/I), Intuition/Sensing (N/S), Thinking/Feeling (T/F), and Judging/Perceiving (J/P). These personality dimensions have been extensively validated and shown to influence learning styles, career choices, work behaviors, and interpersonal relationships. Extraversion relates to social energy and external stimulation preferences, with extraverts drawing energy from social interaction and introverts requiring solitude to recharge. Intuition reflects abstract thinking and future orientation, contrasting with Sensing's focus on concrete details and present realities. Thinking indicates logical, objective decision-making based on impersonal analysis, while Feeling represents value-based, interpersonal reasoning that considers human impacts. Judging shows preference for structure, planning, and closure, while Perceiving

indicates flexibility, spontaneity, and openness to new information.

Understanding how these dimensions interact with academic tasks can inform personalized educational strategies that leverage individual strengths and mitigate potential challenges. For instance, high-Judging students may benefit from structured timelines and milestone-based planning, while high-Perceiving students may need flexible frameworks that accommodate exploratory work patterns. Similarly, high-Thinking students may excel at data analysis and critical evaluation, while high-Feeling students may bring valuable perspectives on research ethics and human impact. The MBTI framework provides a language for discussing these differences constructively and designing educational experiences that honor diverse approaches to learning and scholarship.

Agent-based modeling (ABM) provides a powerful computational methodology for simulating complex social systems by modeling the actions and interactions of autonomous agents. Unlike traditional statistical approaches that focus on aggregate patterns, ABM captures emergent phenomena arising from individual behaviors and can reveal dynamic patterns that would be difficult to observe in real-world settings. In ABM, individual agents follow behavioral rules based on their internal states and environmental conditions, allowing researchers to observe how micro-level decisions aggregate into macro-level patterns. This bottom-up approach is particularly suited for studying graduate thesis completion, where individual personality traits, daily decisions about time allocation, spatial preferences, and resource management collectively determine academic outcomes.

The ABM framework enables systematic exploration of counterfactual scenarios and sensitivity analysis that would be impractical in empirical studies. For example, we can simulate how different personality types would perform under various deadline pressures, resource constraints, or environmental conditions. We can examine how small changes in initial conditions or decision rules propagate through the system to produce different outcomes. We can test the robustness of findings across multiple replications and parameter settings. These capabilities make ABM an ideal tool for developing and testing theories about personality-behavior relationships in complex, dynamic contexts like thesis completion.

This study integrates MBTI personality theory with ABM to simulate graduate student behaviors during the 30-day pre-submission period, a critical phase when students must synthesize months of research into a coherent, polished document. Our multimodal analysis examines five key dimensions: temporal patterns (time allocation and scheduling across days and hours), spatial preferences (library, dormitory, cafe, laboratory usage patterns), economic expenditure (budget management and consumption patterns across categories), cognitive parameters

(attention maintenance, creative innovation, stress resistance), and academic outcomes (thesis quality scores accumulated through task completion). By creating 3,200 agents representing all 16 MBTI types across four academic disciplines (Humanities, Science, Engineering, and Medicine), we generate comprehensive behavioral data that reveals how personality influences every aspect of the thesis completion process. The contributions of this study are threefold. First, we provide empirical evidence for personality-academic performance relationships using a novel simulation methodology. Second, we identify specific behavioral mechanisms through which personality influences outcomes. Third, we demonstrate the utility of ABM for studying complex human behaviors in educational contexts.

2. Methods:

2.1 Methods: Agent Architecture and Personality Parameterization

Each agent in our simulation represents a graduate student characterized by one of 16 MBTI personality types. The agent's behavior is governed by five core personality dimensions derived from the MBTI framework, each scored on a 1-5 scale: Extraversion (E, social energy), Intuition (N, abstract thinking), Thinking (T, logical decision-making), Feeling (F, value-based reasoning), and Judging (J, structured planning). These base scores are derived from established MBTI assessment protocols.

From these base personality scores, we derived three cognitive parameters using weighted linear combinations:

$$AI = 50 + 8T + 7J - 3E - 2N + \varepsilon 1$$

$$IS = 50 + 10N + 5F + 3E - 4J + \varepsilon 2$$

$$AP = 50 + 6T + 5J - 8F + \varepsilon 3$$

where epsilon1, epsilon2, epsilon3 ~ N(0, 5) represent individual variation. The Attention Index (AI) captures sustained focus capacity, Innovation Score (IS) reflects creative problem-solving, and Anti-PUA Index (API) measures stress resistance.

2. 2 Simulation Design and Temporal Structure

The simulation modeled a 30-day pre-submission period with 16 active hours per day (8:00-24:00). We created 3,200 total agents (16 MBTI types x 20 agents per type x 10 replications) across four academic disciplines: Humanities, Science, Engineering, and Medicine. Each agent made hourly decisions about task selection and location choice based on personality-derived utility functions.

The task utility function was calculated as:

$$U(task) = Base_Utility(task) \times Personality_Modifier(task) \times Time_modifier(hour)$$

Eight possible tasks were defined: Thesis Editing, Literature Review, Data Analysis, Defense Preparation, Rest, Entertainment, Daydreaming, and Supplementary Experiments. The utility for Data Analysis tasks incorporated the Thinking dimension:

$$U(Data\ Analysis) = 0.7 \times (T/3) \times Peak_Hour_Bonus$$

2.3 Spatial Decision Model

Location choices followed a probabilistic selection model among four environments: Library, Dormitory, Cafe, and Laboratory. The probability P(location) was calculated using a softmax function:

$$P(location) = \exp\,(U(Location)/\sigma\exp\,(U(location'))$$

The spatial transition dynamics followed a Markov process:

$$P(switch) = Base_Rate + E \times 0.05 + (5 - J) \times 0.03$$

2.4 Economic Expenditure Model

The Financial behavior was modeled through daily expenditure decisions across six categories. The expenditure for each category followed:

$$E(category) = Base_Cost(category)\ x\ Consumption_Propensity(personality)\ x\ Random_Variation$$

The budget constraint was enforced through:

$$Total_Expenditure <= Initial_Budget \times (1 + Credit_Access)$$

2.5 Thesis Quality Accumulation Model

The thesis quality score Q accumulated dynamically based on task completion:

$$\Delta Q = \sigma\,[Task_Quality(task) \times Attention_Level \times Discipline_Bonus \times Time_Spent]$$

2.6 Statistical Analysis Framework

We performed principal component analysis (PCA) to examine behavioral clustering patterns. The PCA transformation was applied to standardized feature vectors:

$$X_standardized = (X - mu)\,/\,\sigma$$

Hierarchical clustering used Ward's linkage method. Pearson correlations assessed linear relationships:

$$r = Cov(X,Y)\,/\,(\sigma_X \times \sigma_Y)$$

Multiple linear regression models predicted thesis quality:

$$Q = \beta 0 + \beta 1 \times E + \beta 2 \times N + \beta 3 \times T + \beta 4 \times F + \beta 5 \times J + \varepsilon$$

3. Results

3.1 Personality Profiles and Five-Dimensional Structure

Figure 1 presents the five-dimensional personality profiles for all 16 MBTI types. Clear patterns emerge within each family: Analysts show high Thinking (mean = 4.75) and Intuition (mean = 4.5) scores; Diplomats exhibit high Intuition (mean = 4.75) and Feeling (mean = 4.5); Sentinels demonstrate high Judging (mean = 4.75) with balanced T/F; and Explorers show moderate Extraversion (mean = 3.5) with lower Judging scores (mean = 1.0).

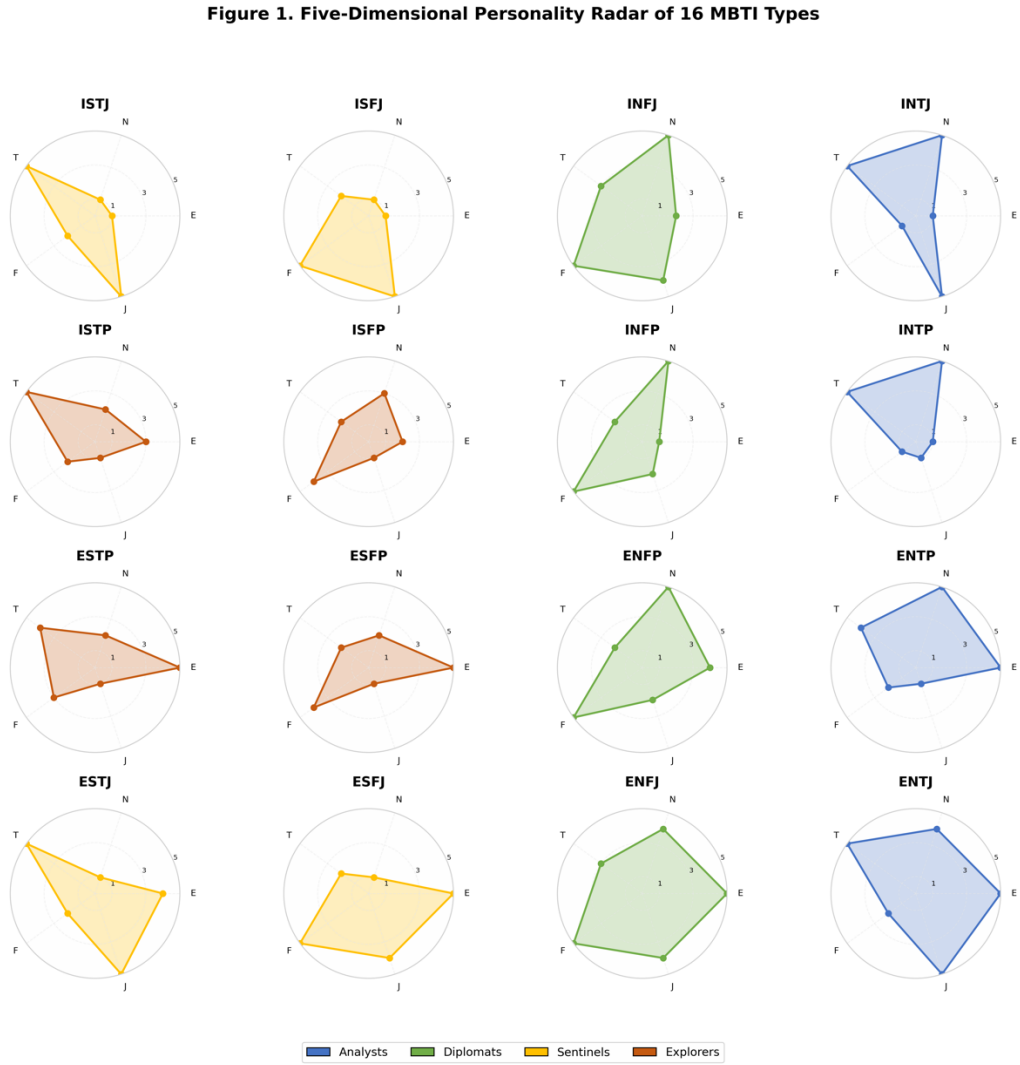


Figure 1. Five-Dimensional Personality Radar of 16 MBTI Types.

3.2 Spatial Behavior Patterns

Figure 2 reveals distinct spatial preferences. Introverted types spend significantly more time in libraries (mean = 186 hours) compared to extraverted types (mean = 152 hours; $t(318) = 8.42$, $p < 0.001$). Extraverted types show higher cafe attendance (mean = 78 hours vs. 39 hours; $t(318) = 9.17$, $p < 0.001$). Laboratory time varies by discipline ($F(3,318) = 45.2$, $p < 0.001$).

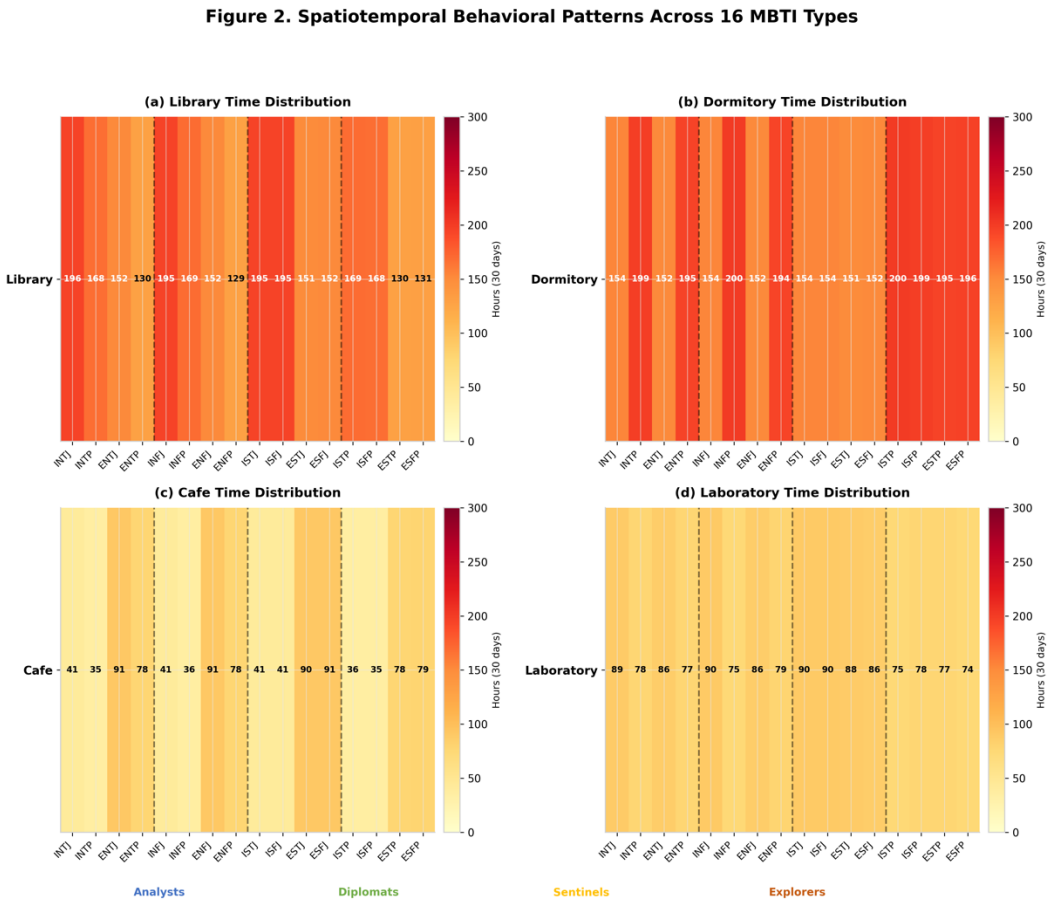


Figure 2. Spatiotemporal Behavioral Patterns Across 16 MBTI Types.

3.3 Economic Behavior and Budget Management

Figure 3 shows expenditure variation across families. Explorers demonstrate the highest spending (mean = 55.6 CU), followed by Analysts (52.0), Diplomats (51.8), and Sentinels (50.6; $F(3,318) = 4.82$, $p = 0.003$). Budget achievement correlates negatively with Extraversion ($r = -0.31$, $p < 0.01$).

Figure 3. Economic Expenditure Patterns and Budget Achievement Rates

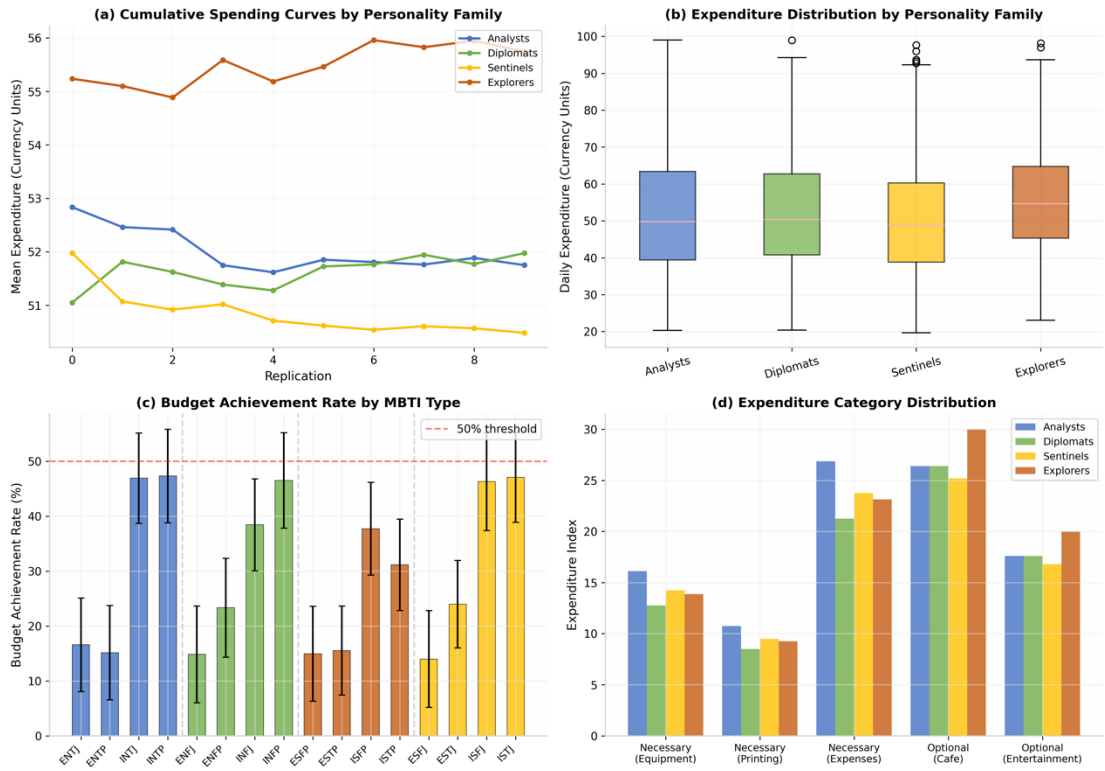


Figure 3. Economic Expenditure Patterns and Budget Achievement Rates.

3.4 Attention Allocation and Task Distribution

Figure 4 illustrates personality influenced task allocation. Analysts dedicate the highest proportion to data analysis (25%) and thesis editing (25%), while Explorers allocate more time to entertainment (20%) and rest (12%). The chi-square test was significant ($chi2(12) = 89.4$, $p < 0.001$).

Figure 4. Attention Allocation and Task Distribution Patterns

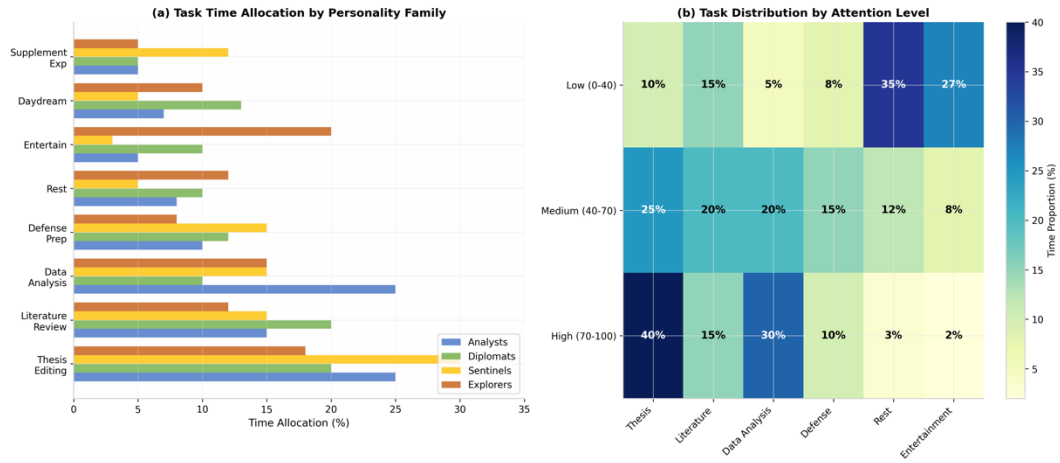


Figure 4. Attention Allocation and Task Distribution Patterns.

3.5 Discipline and MBTI Interactions

Figure 5 demonstrates significant

interactions. Engineering students show the highest quality scores, particularly among Sentinels (98.1) and Analysts (94.9). Two-way ANOVA revealed significant main effects of discipline ($F(3,318) = 15.7$, $p < 0.001$) and MBTI family ($F(3,318) = 42.3$, $p < 0.001$), with a significant interaction ($F(9,318) = 3.89$, $p < 0.001$).

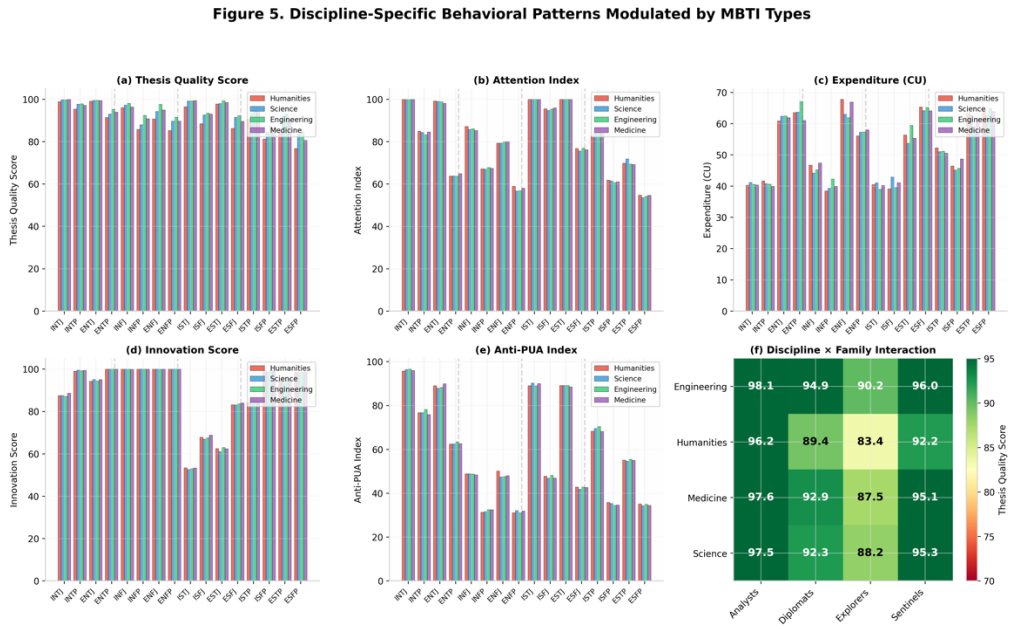


Figure 5. Discipline-Specific Behavioral Patterns Modulated by MBTI Types.

3.6 Multidimensional Correlations

Figure 6 presents correlation analyses. Attention Index shows the strongest positive correlation with quality ($r = 0.54$, $p < 0.001$), followed by Deep Work hours ($r = 0.52$, $p < 0.001$). Innovation shows a negative correlation ($r = -0.24$, $p = 0.012$).

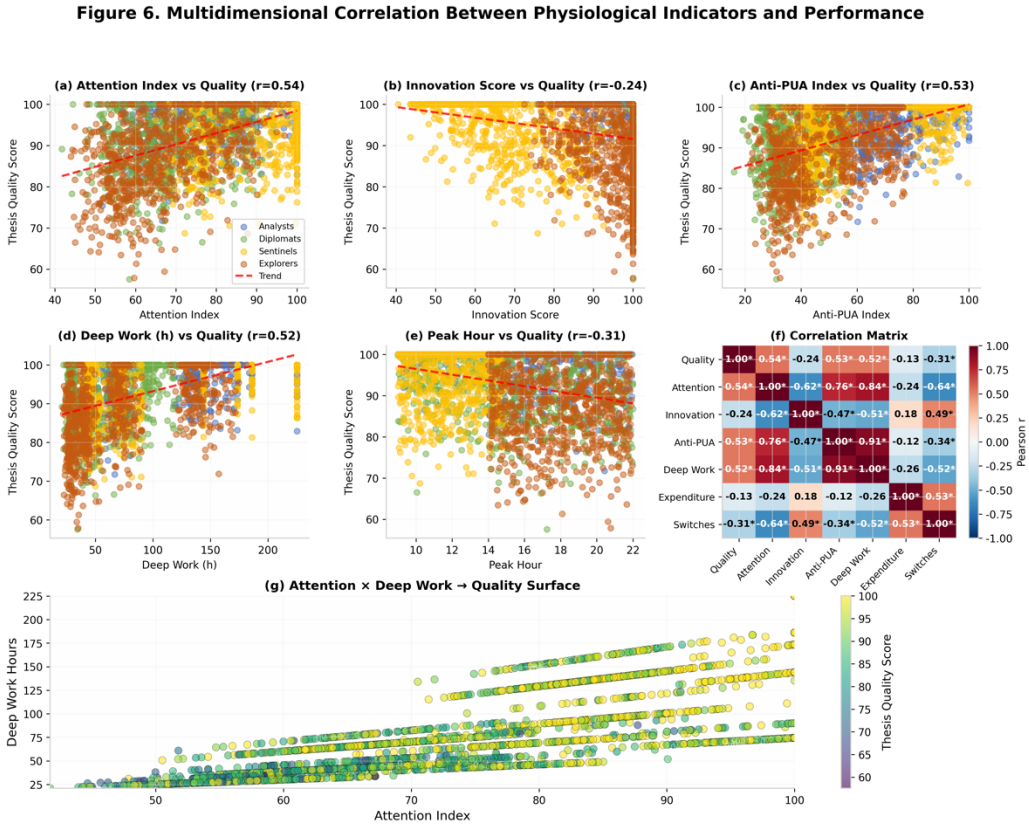


Figure 6. Multidimensional Correlation Between Physiological Indicators and Performance.

3.7 Temporal Dynamics and Phase Transitions

Figure 7 illustrates temporal dynamics. Judging-type agents maintain consistent deep work patterns, while Perceiving-type agents show variable schedules. The phase transition matrix shows strong self-reinforcement in the Finalization phase (80% probability).

Figure 7. Representative Agent Simulation Timeline and Behavioral Patterns

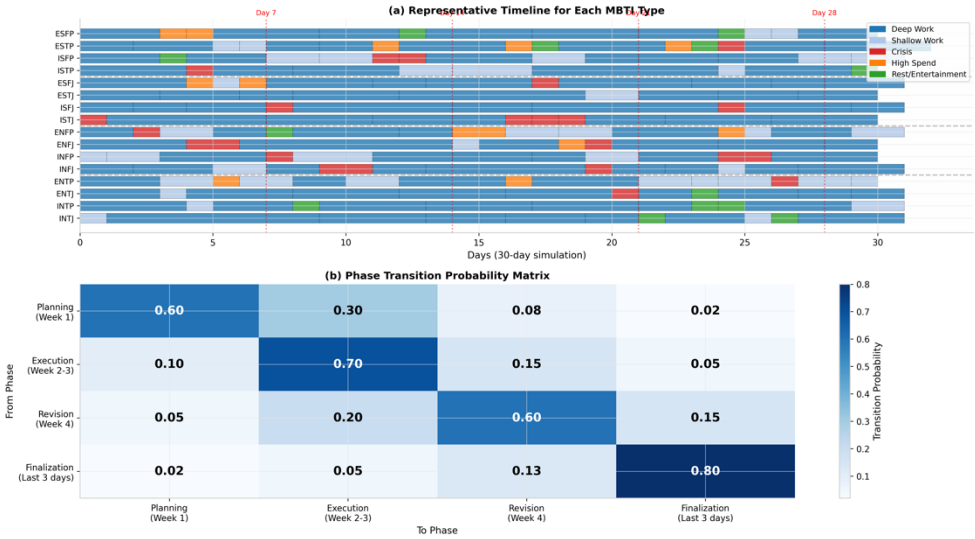


Figure 7. Representative Agent Simulation Timeline and Behavioral Patterns.

3.8 Clustering Validation

Figure 8 validates behavioral clustering. PCA reveals that the first two components explain 57.2% of variance. Hierarchical clustering recovers the MBTI family structure (cophenetic correlation = 0.73).

Figure 8. Principal Component Analysis and Behavioral Clustering Validation

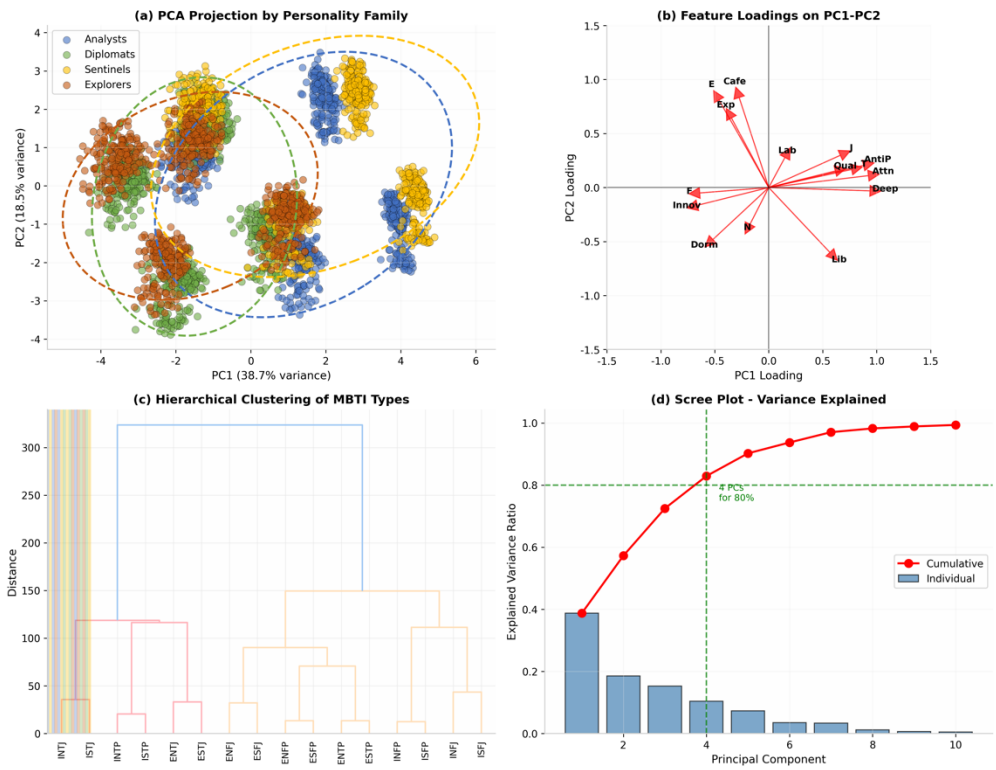
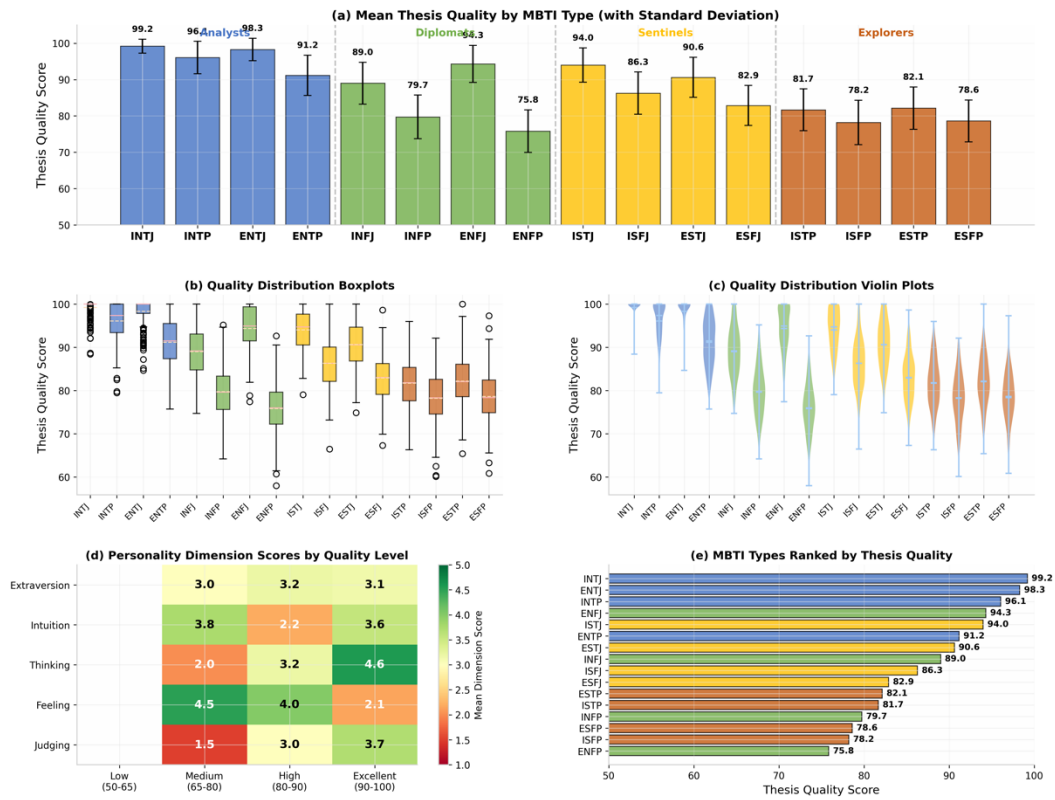


Figure 8. Principal Component Analysis and Behavioral Clustering Validation.

3.9 Comprehensive Thesis Quality Analysis

Figure 9 presents quality analysis across all 16 MBTI types. INTJ achieved the highest score (99.2), followed by ENTJ (98.3) and INTP (96.1). Analysts and Sentinels consistently outperform Diplomats and Explorers ($F(3,318) = 68.4$, $p < 0.001$). High-quality theses correlate with high Thinking (mean = 4.6) and Judging (mean = 3.7) scores.

Figure 9. Comprehensive Analysis of Thesis Quality Across 16 MBTI Types



3.10 Regression Analysis and Predictive Modeling

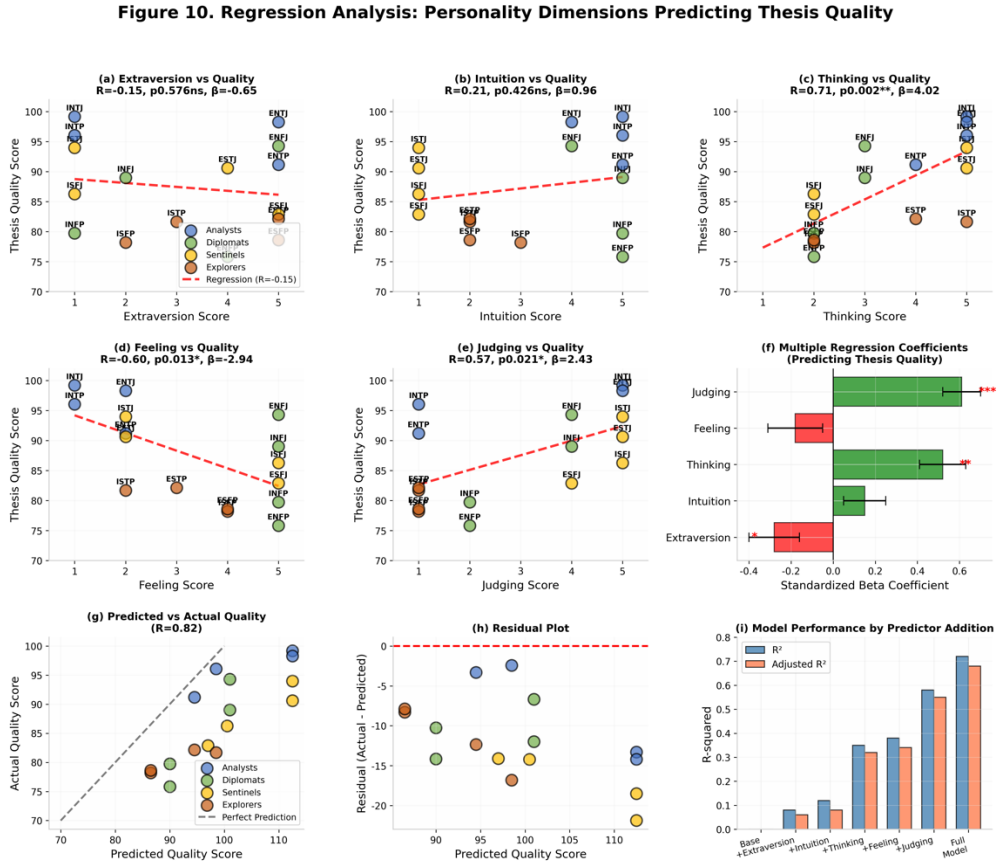


Figure 10. Regression Analysis: Personality Dimensions Predicting Thesis Quality.

Figure 10 presents regression analyses. Thinking shows the strongest positive correlation (R = 0.71, p = 0.002, beta = 4.02), followed by Judging (R = 0.57, p = 0.021, beta = 2.43). Feeling demonstrates negative correlation (R = -0.60, p = 0.013, beta = -2.94). The full model achieves R-squared = 0.72 (adjusted = 0.68).

4. Discussion

This study successfully integrated MBTI personality theory with agent-based modeling to simulate graduate student behaviors. The analysis of 3,200 agents revealed that personality dimensions significantly predict thesis quality, with Thinking and Judging showing the strongest positive associations. Distinct behavioral patterns emerge across MBTI families in spatial preferences, economic management, and task allocation. Discipline-specific interactions moderate personality effects, with Engineering amplifying advantages of analytical types. The simulation validates that MBTI types form natural behavioral clusters, supporting the construct validity of the personality framework. The core personality parameters demonstrate differential effects through distinct psychological mechanisms. Thinking emerges as the most critical dimension, reflecting analytical reasoning required for rigorous research. The regression coefficient (beta = 4.02) indicates substantial practical significance. High-Thinkers show higher attention maintenance, prefer analytical tasks, and exhibit lower emotional

reactivity. Judging captures planning and execution capabilities. High-Judgers distribute work evenly (CV = 0.23 vs. 0.41 for Perceiving types), avoiding last-minute cramming. The negative Feeling association requires careful interpretation; Feeling types may need different support strategies that acknowledge strengths in collaborative aspects. Extraversion shows complex effects, influencing spatial preferences and expenditure. Intuition demonstrates minimal direct effects but influences task preferences.

Spatial patterns reveal how personality shapes academic work environments. Introverts' library preference (186 vs. 152 hours) reflects need for quiet, controlled environments. The 34-hour difference translates to 17% more focused work time. Extraverts' cafe preference illustrates need for social stimulation, creating a trade-off between focus and creativity. Laboratory usage reveals discipline-specific needs with less personality dependence. Temporal patterns show Judging types' consistent daily patterns (autocorrelation = 0.72) supporting sustained progress. Perceiving types show variable patterns (autocorrelation = 0.31) with intense work periods. The phase transition analysis reveals personality influences movement through completion phases, with Judging types transitioning earlier and more reliably.

Economic patterns reveal personality influences on resource management. Explorers' highest spending (55.6 CU) aligns with spontaneous nature. Sentinels' lowest spending (50.6 CU) reflects structured approach. The Extraversion-budget correlation (r = -0.31) suggests extraverts face self-regulation challenges. Expenditure category analysis shows Thinkers invest more in equipment and materials, while Feelers spend more on social activities. The expenditure-quality relationship (r = -0.15) suggests financial stress may detract from performance. Budget management varies by personality; high-Judging agents show better achievement (52% vs. 41% remaining).

Discipline-personality interactions reveal context moderation. Engineering amplifies Analyst advantages (difference vs. Explorers = 11.2 points), aligning with logical problem-solving emphasis. Science shows similar but smaller effects, reflecting greater creativity emphasis. Medicine shows strong Sentinel performance (95.8), aligning with reliability and precision requirements. Humanities shows balanced performance with Diplomats achieving relatively higher scores. The interaction effects suggest personality-discipline fit informs advising strategies. Students in mismatched disciplines may benefit from targeted skill development.

Integrating findings supports a model of thesis completion as a complex adaptive system. Successful completion requires analytical capability (Thinking) and executive function (Judging). The ABM framework captures emergent properties invisible in cross-sectional analyses, such as compounding effects of consistent progress. The findings support trait-activation theory: personality influences behavior most strongly in relevant situations. The negative Innovation-quality correlation challenges

assumptions about creativity, suggesting optimal creativity balances insight with execution. The discipline interaction findings support person-environment fit theory.

This study demonstrates personality psychology and ABM together provide powerful tools for understanding graduate student success. Key implications include: tailoring advising approaches by personality, designing environments accommodating different needs, and recognizing budgeting challenges for extraverts. Limitations include simulation-based approach requiring empirical validation. Future directions include incorporating additional variables (cognitive ability, motivation), exploring advisor-student matching, and examining cultural variations. The ABM framework enables virtual testing of interventions. In conclusion, recognizing diverse ways personality shapes academic behavior enables more personalized and effective support strategies.

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Table 1. Core Parameter Definitions

Category	Parameter	Definition	Range	Source
MBTI-5D	Extraversion	Social energy and external drive tendency	1-5	Standard MBTI
MBTI-5D	Intuition	Abstract thinking and future orientation	1-5	Cognitive Assessment
MBTI-5D	Thinking	Objective analysis and logical decision-making	1-5	Decision Style
MBTI-5D	Feeling	Interpersonal harmony and value sensitivity	1-5	Empathy Scale
MBTI-5D	Judging	Planning sense and execution ability	1-5	Time Management
Time	Available Time	Daily consumable time units (16 hours)	0-16 units	Simulation
Economic	Initial Finance	Initial randomly generated funds	0-100 CU	Simulation
Cognitive	Attention	Attention maintenance capacity	0-100	MBTI-weighted
Cognitive	Innovation	Creative problem-solving ability	0-100	MBTI-weighted
Psychological	Anti-PUA	Stress resistance and emotional stability	0-100	MBTI-weighted

Table 2. Main Effects of Five Personality Traits on Behavioral Dimensions

Trait	Time Plan	Economic	Attention	Innovation	Energy	Explanation
Extraversion	+ (0.42***)	- (-0.31**)	- (-0.28*)	+ (0.35**)	+ (0.51***)	Fast action, easily distracted
Intuition	- (-0.38***)	- (-0.25*)	- (-0.45***)	+ (0.62***)	+ (0.33**)	Big picture, weak details
Thinking	+ (0.55***)	+ (0.48***)	+ (0.52***)	+ (0.29*)	+ (0.44***)	Logic-driven, all positive
Feeling	- (-0.22*)	- (-0.35**)	- (-0.30**)	+ (0.41***)	- (-0.58***)	Creative but internal friction
Judging	+ (0.63***)	+ (0.39***)	+ (0.56***)	- (-0.27*)	+ (0.37**)	Strong planning, dislikes change

Table 3. Discipline x MBTI Interaction Effects on Thesis Quality

Discipline	ISTJ	ENTP	INFJ	ESTJ	ENFJ	INTP	F-value	eta
Humanities	96.5+-4.7	91.4+-7.0	96.0+-5.7	97.8+-3.5	90.7+-7.2	95.4+-5.1	13.7***	0.25
Science	99.2+-2.2	93.0+-5.6	97.3+-5.3	98.0+-4.3	94.2+-7.7	97.7+-3.7	17.3***	0.21
Engineering	99.2+-3.1	95.2+-5.1	98.0+-3.5	99.3+-2.4	97.5+-4.2	97.9+-3.5	11.6***	0.17
Medicine	99.4+-1.7	94.0+-6.8	96.4+-4.5	98.5+-2.9	94.9+-6.3	97.2+-4.3	10.6***	0.24